



GEOSPATIAL ELECTRIFICATION SCENARIO MODELLING FOR ZAMBIA USING ONSSET



Brian SIAKWEENDA – siakweendabrian@gmail.com
 Rejoice MUTUTSA – rejoicemututsa23@gmail.com

Energy Modelling Platform Africa (EMP-A)

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BACKGROUND

The Demography, Generation Mix, and Access rates are indicated below:

01



Demography

- Population of 19.6 Million - 2022
- 3.1% Growth Rate
- Population of 25 Million by 2030
- 40% Urban Ratio
- 60% Rural Ratio

02



Generation Mix

- 3,965.8 MW Installed Capacity
- Average Demand: 2,450.7MW
- 83% Hydro
- 8.7% Coal
- 3.2% Solar
- 2.9% HFO
- 2.2% Diesel

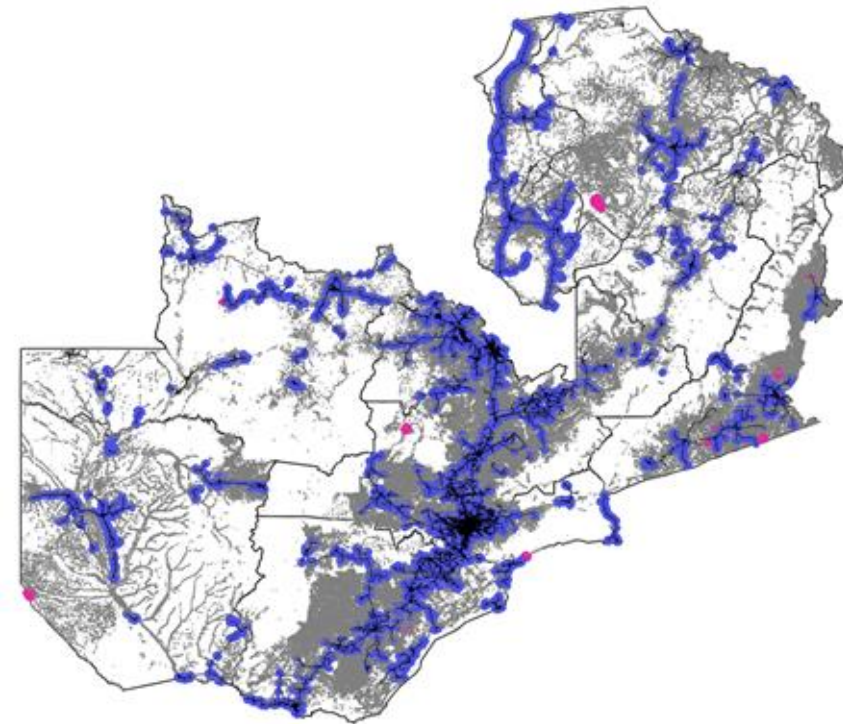
03



Access Rate

- 53.6% National Access Rate
- 80.3% Access in Urban Areas
- 34% Access in Rural Areas
- Grid Access – 34%
- Off-Grid Access – 19.6

ZAMBIA MAP LAYOUT



Legend

- existing_mv
- Existing grid electrified
- Mini-Grid Electrified
- Unelectrified clusters
- Admin Boundaries

0 100 200 km

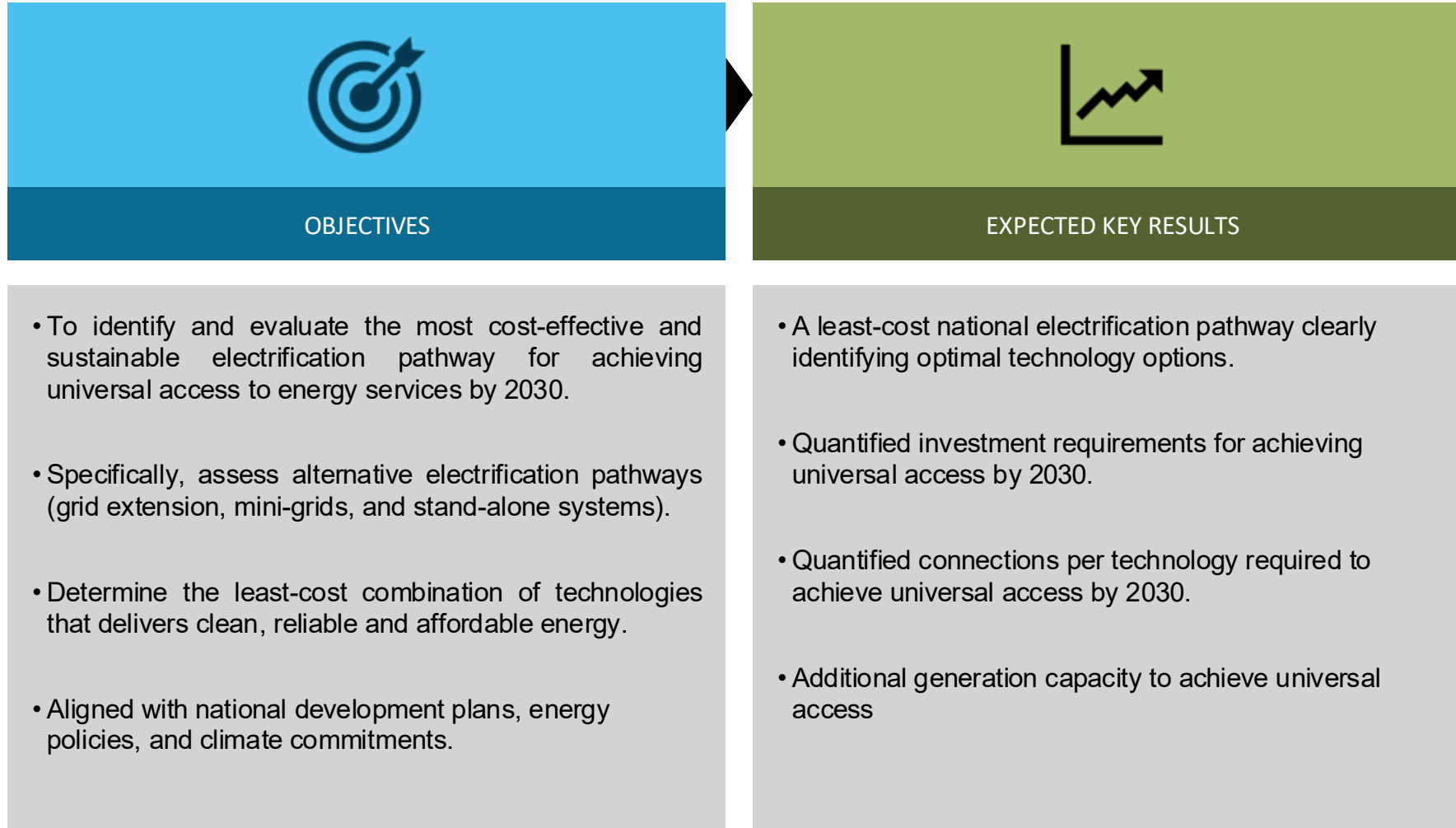
BACKGROUND CONT...

The Key Challenges, Access Targets and Strategic Targets are as Highlighted below:

1	KEY CHALLENGES	❑ Low Energy Access – 46.4% of the Population has no access to electricity ❑ Infrastructure Deficiency: Insufficient to meet emerging demand ❑ Dependence of Hydropower Generation: High Risk to Climate Change Effects. ❑ High Connection Fee: Rural areas remain unconnected	
2	ACCESS TARGETS	❑ Achieve Universal Access (100%) to electricity by 2030 ❑ Achieved through Grid and Off-grid solutions	
3	STRATEGIC TARGETS	❑ NATIONAL ENERGY POLICY, IRP & ZAMBIA ENERGY COMPACT ▪ Increase the share of non-hydro renewable energy in the energy mix from 3% to 33% by 2030 ▪ Increase the number of Grid connections from 60,000 in 2022 to 120,000 by 2030	

OBJECTIVES

The Objects and Expected Key Results are outlined below:



ELECTRIFICATION SCENARIOS CONSIDERED

1. TOP-DOWN LOW – GRID UNCAPPED

Demography

- ❑ Start Year Population (2024): 20.8 Million
- ❑ Annual Growth Rate: 3.1%
- ❑ End Year Population (2030): 25 Million
- ❑ Urban Ratio: 40%

Modelling Parameters

- ❑ Yearly Grid Connections: Not Capped
- ❑ Maximum Grid Int. Cost: \$1,300/HH
- ❑ Grid Generation Cost: \$0.075/kWh
- ❑ Grid Losses Factor: 0.15
- ❑ Service connection: \$230/connection
- ❑ Service Transf. Type: 50kVA
- ❑ MV Network: 11 & 33kV
- ❑ MV network cost: \$18,862/km
- ❑ Max Grid Extension Dist. 50km

Target Electrification Access

- ❑ 100% by 2030 in both urban and rural areas

Vs



2. TOP-DOWN LOW – GRID CAPPED

Demography

- ❑ Start Year Population (2024): 20.8 Million
- ❑ Annual Growth Rate: 3.1%
- ❑ End Year Population (2030): 25 Million
- ❑ Urban Ratio: 40%

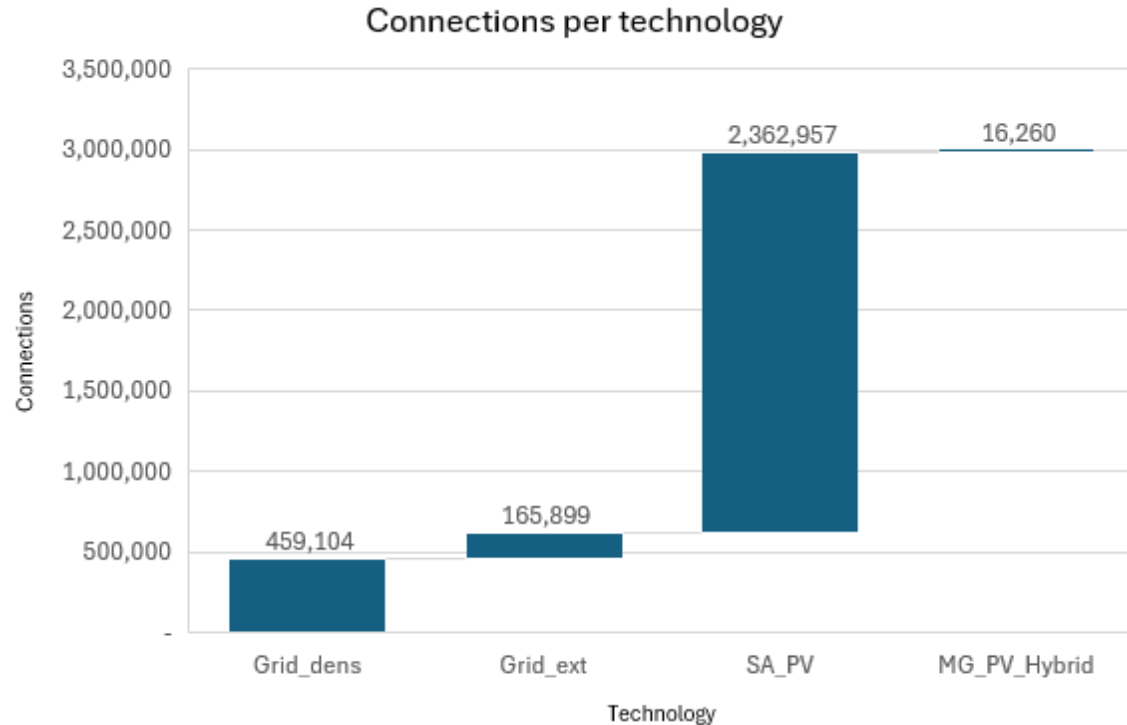
Modelling Parameters

- ❑ Yearly Grid Connection: 100,000
- ❑ Maximum Grid Int. Cost: Determined by Model
- ❑ Minimum HH Size for Minigrid: 100
- ❑ Minigrid Consideration Radius: 1km
- ❑ PV & Inverter Cost: \$1,292/kW
- ❑ Battery Cost: \$481/kWh
- ❑ Demand Targets:
 - Tier 1 - Low rural (20 kwh)
 - Tier 2 - High rural (73 kwh)
 - Tier 4 – Urban (1,241 kwh)

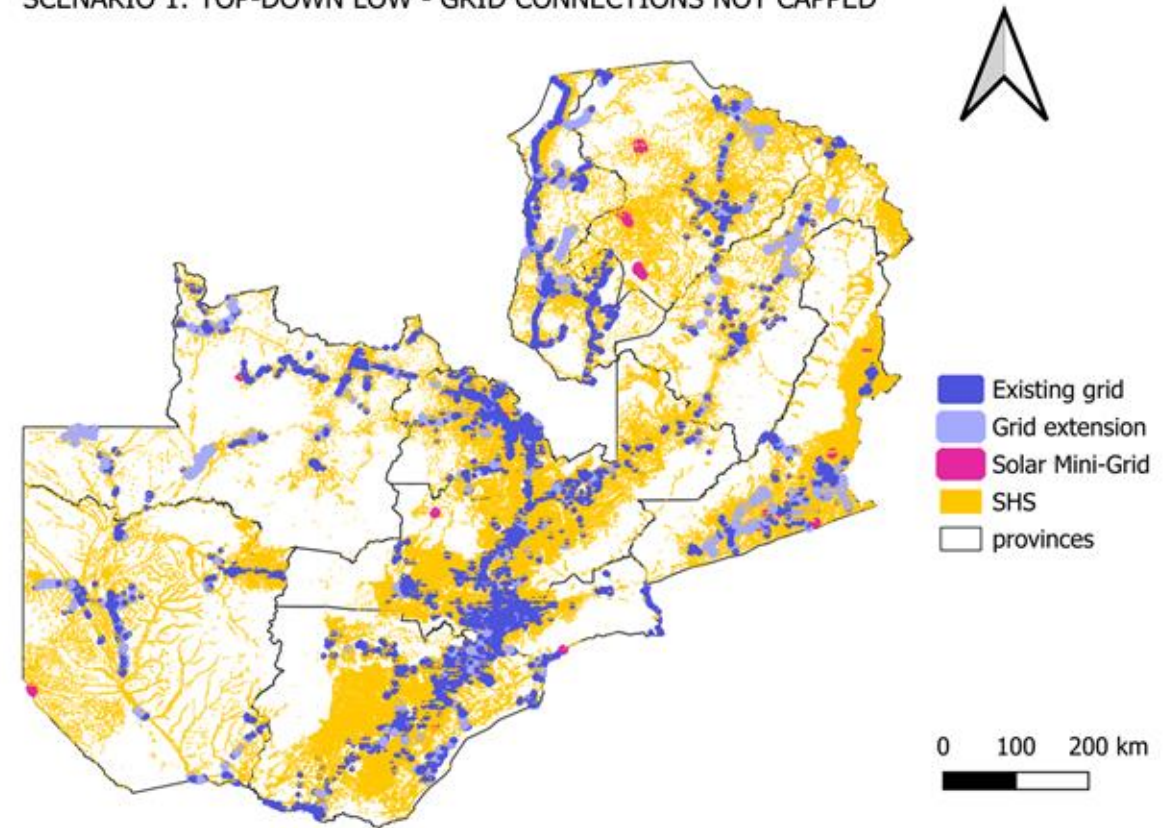
Target Electrification Access

- ❑ 100% by 2030 in both urban and rural areas

RESULTS: SCENARIO 1 – GRID CONNECTION NOT CAPPED

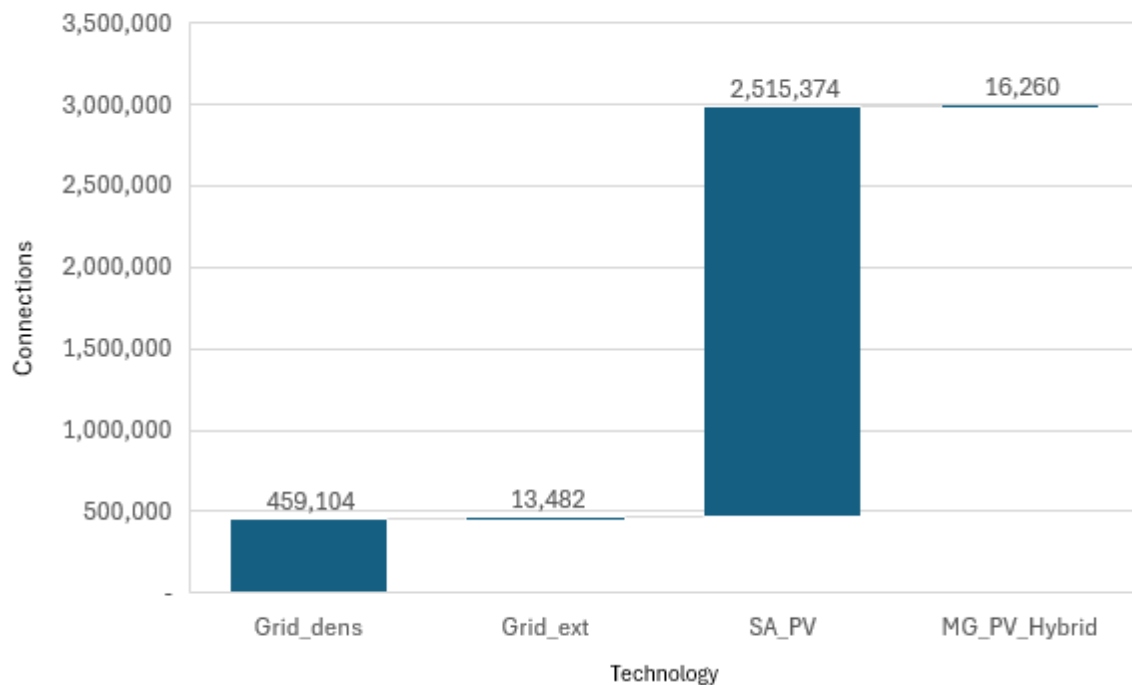


SCENARIO 1: TOP-DOWN LOW - GRID CONNECTIONS NOT CAPPED

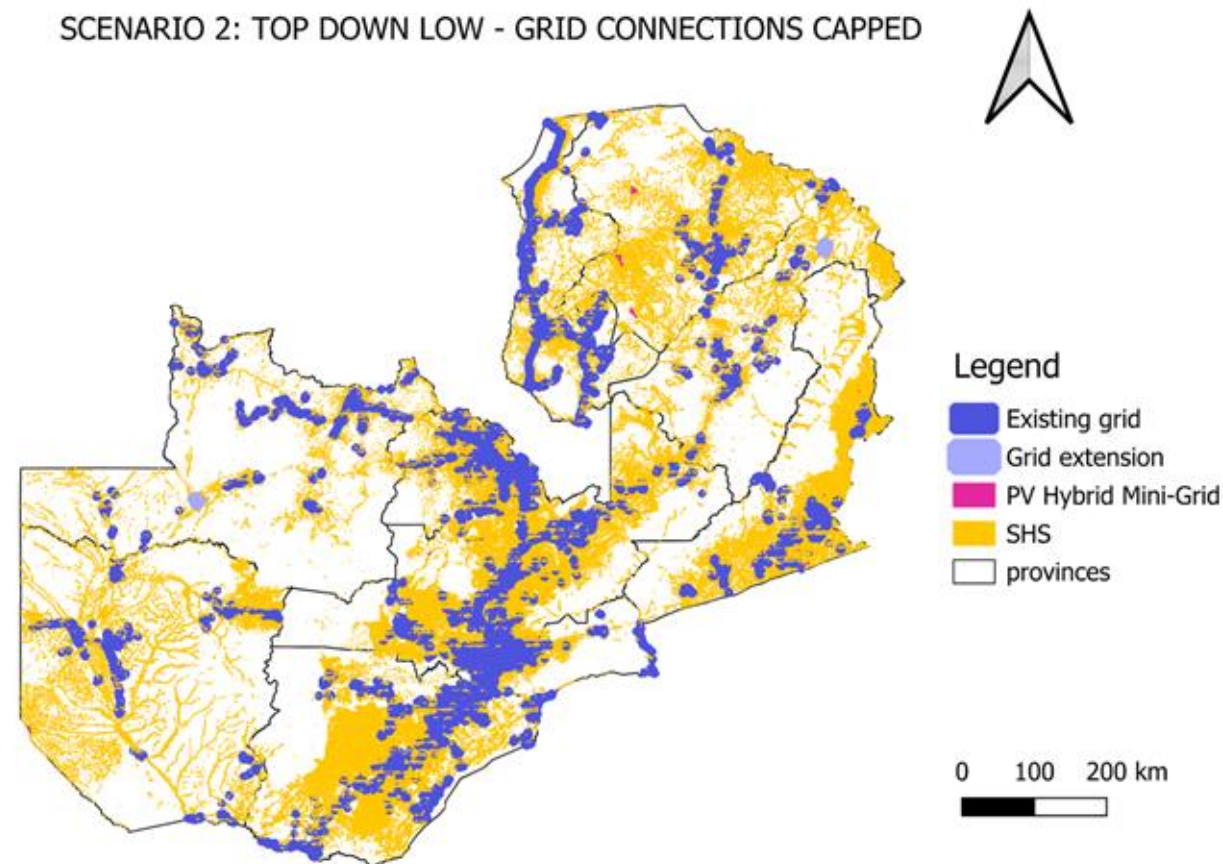


RESULTS: SCENARIO 2 – GRID CONNECTION CAPPED

Connections per technology



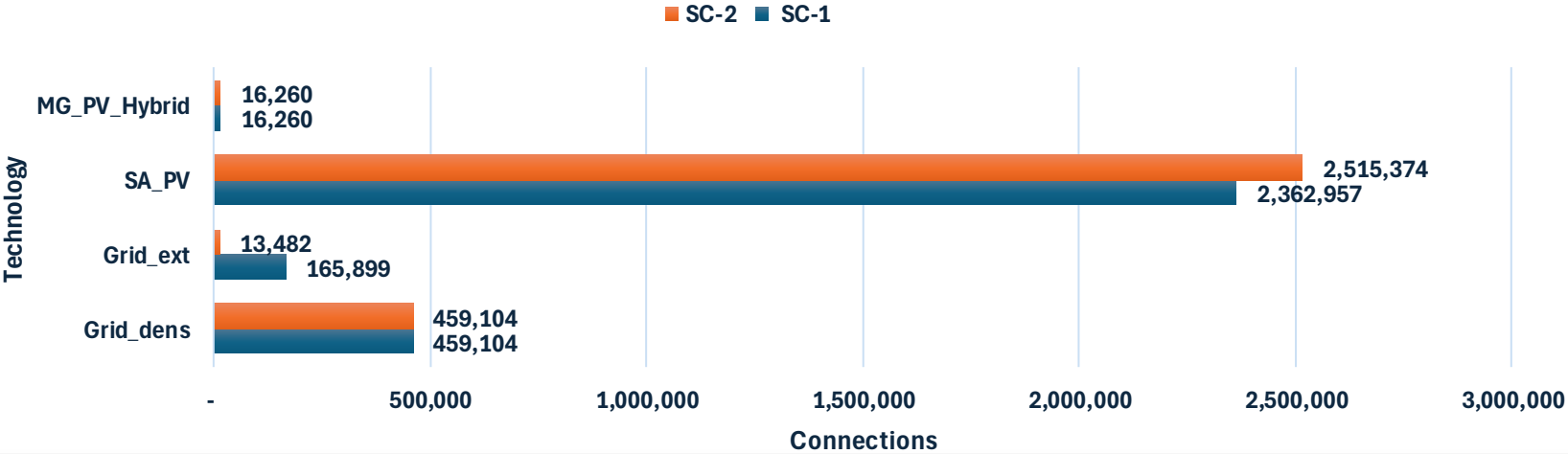
SCENARIO 2: TOP DOWN LOW - GRID CONNECTIONS CAPPED



SCENARIO COMPARISON

No.	DESCRIPTION	SCENARIO 1	SCENARIO 2
1.	Number of New Connections	3,004,220	3,004,220
2.	Addictional Capacity (MW)	169.95	171.68
3.	Investment (Million USD)	1,448.19	1373.42
4.	Emissions (tCO2)	78.83	76.87
5.	Electrification Technologies	- Grid Densification - Grid Extension - Solar Home Systems - Solar Mini grids	- Grid Densification - Grid Extension - Solar Home Systems - Solar Mini grids

Scenario Connections Comparison



CONCLUSION AND POLICY INSIGHT

- The scenario analysis indicates that achieving universal access to electricity by 2030 will require a diversified mix of technologies
- **Scenario 1** is well suited for public sector investment, as it prioritizes a higher number of grid extension projects.
- **Scenario 2**, which incorporates an increased share of connections through solar home systems, is more attractive for private sector participation.
- **Scenario 1** emerges as the most effective pathway for achieving universal access, as it strategically proposes grid extensions in areas where demand clusters are close to existing infrastructure, thereby making grid expansion both economically and technically viable.

Future Work

- Integration of an attribute allocation algorithm to systematically assign technical parameters to model outputs.
- Development of a routing algorithm to optimise network layouts for recommended off-grid systems.
- Incorporation of solar home systems (SHS) into the model to contribute to electricity access rates within the overall modelling framework.